## $^{37}$Cl(p,t) 1971Vi02,2017Ch22

$^{37}Cl \rightarrow ^{35}Cl$+2n from $J^{\pi}=3/2^{+}$ $^{37}$Cl ground state.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. Targets were natural purified chlorine gas 100 mm in diameter and 25 mm in height and solid AgCl of isotopic $^{37}$Cl. The reaction particles were detected using two separate counter telescopes consisting of a 200-$\mu$m phosphorous-drifted silicon $\Delta$E detector, a 3-mm lithium-drifted silicon E detector, and a 3-mm lithium-drifted silicon E-reject detector with FWHM=130 keV for tritons. Measured $\sigma(E_t,\theta)$. Deduced L-transfers for 7 levels from zero-range Nelson spin-dependent-DWBA analysis of the measured $\sigma(\theta)$.

2017Ch22: a 30-MeV proton beam was produced from the Holifield Radioactive Ion Beam Facility at Oak Ridge National Laboratory. The target was ~600 $\mu$g $PbCl_2$ (~95% enriched and ~160 $\mu g/cm^2$ in $^{37}$Cl) on a 20 $\mu g/cm^2$ parylene ($C_8H_8$) backing. Tritons were detected using the annular silicon detector array (SIDAR) consisting of 100-$\mu$m-$\Delta$E and 1000-$\mu$m-E telescopes with FWHM=60-100 keV. Measured $\sigma(E_t,\theta)$. Deduced levels, J, $\pi$, and L-transfers from TWOFNR-DWBA analysis of the measured $\sigma(\theta)$. Provided the first spin-parity constraint of a 306-keV $^{34}S(p,\gamma)$ resonance potentially important for nova nucleosynthesis.

Other: 1985Mi06.

### $^{35}$Cl Levels

| E(level)† | L‡ | Comments |
|---|---|---|
| 0 | 0 | L: from 2017Ch22. Other: 0+2 in 1971Vi02.<br>Relative peak intensity=100 (1971Vi02). |
| 1220 *40* | | Relative peak intensity=0 (1971Vi02). |
| 1750 *40* | 2 | Relative peak intensity=41 (1971Vi02). |
| 2650 *40* | 2 | Relative peak intensity=41 (1971Vi02). Unresolved 2645 and 2694 doublet. |
| 3000 *40* | | Relative peak intensity=5 (1971Vi02). |
| 5650 *50* | 0+2 | Relative peak intensity=22 (1971Vi02). |
| 6677 *15* | (2) | E(level): from 2017Ch22. 2017Ch22 adopts L=2 and states that combinations of L=0+2 or L=1+3 are also possible. Because L=2 and L=0 are better fits than L=1 and L=3, a positive parity assignment for the 6677 level is strong. |
| 7250 *50* | | Relative peak intensity=9 (1971Vi02). |

† From 1971Vi02, unless otherwise noted.
‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1971Vi02, unless otherwise noted.